Step 2 Research Review

What makes a business process worth automating?

Evidence base for automation-opportunity prioritisation from desktop-operation telemetry.

Before prioritising automation opportunities in Dataset B, we reviewed research, industry practice and Japanese public- and private-sector evidence to establish which characteristics generally make a business process a strong automation candidate. We then mapped those criteria onto the evidence our telemetry can actually produce.

This document is the *evidence base* for the Step 2 ranking. It does not contain the ranking, and it does not select a candidate. Scope was deliberately bounded to roughly half a day; it is not a systematic review, and where evidence is thin we say so rather than filling the gap.

CONTEXT

Intern selection task: *From operation logs to an automation proposal*. Step 1 (execution recovery from telemetry) is frozen. Step 2 asks which processes should be automated, in priority order, with reasoning.

STATUS

Research artifact. Source of truth for the substance is `docs/STEP2_RESEARCH_REVIEW.md`.

EVIDENCE TAGS USED THROUGHOUT

LIT	Peer-reviewed or scholarly literature.
JP	Japanese government, corporate or industry evidence.
VENDOR	Vendor primary documentation (product documentation, not marketing).
DATA	Measured in our own project data.
INF	Our inference from combining the above.

These categories are never blurred. Tags are legible in grayscale: the letters carry the distinction, the tint is secondary.

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What the research established

Six findings. Each is carried into the Step 2 design; none is new here — they are the conclusions of the review that follows.

01 THE FAILURE MODE

Choosing the wrong process is the leading *documented* cause of automation project failure — reported independently by two systematic reviews. The ordering we propose must therefore rest on characteristics evidence links to success. LIT

02 FREQUENCY SIZES THE PRIZE

Volume measures how large the opportunity is, not whether the work can be automated. Japan's MIC criteria list volume once against four work-property criteria; UiPath's own candidate score omits volume entirely. LIT JP VENDOR

03 DETERMINISM IS PIVOTAL

Bosco et al. define automatability *over UI logs*: an activation condition plus action parameters that are constant or derivable. It is the one criterion in the literature computable from telemetry rather than elicited from experts. LIT

04 JAPANESE GUIDANCE AGREES

The MIC (総務省) RPA guidebook for local government emphasises routine work needing no complex judgement, large repetitive volume, an established *electronic* input source, and a *stable* processing target. JP

05 OUR TELEMETRY COVERS ONE SIDE

We can measure frequency, effort, people, systems, interaction composition, variants and repeatability. We cannot establish labour cost, business criticality, API availability, true volume, error rates or compliance exposure. DATA INF

06 THEREFORE: SEPARATE THE AXES

Step 2 should gate on suitability, rank on impact, present the two as a matrix rather than one score, and surface missing evidence as explicit client validation questions. INF

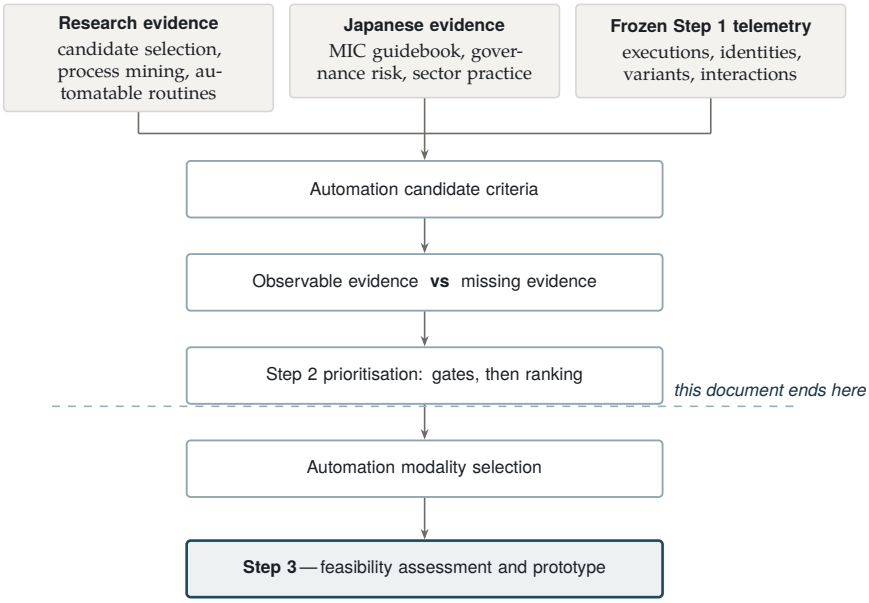


Figure 1. How this research artifact connects to the project. The review supplies the criteria and the observable/missing split; the Step 2 analysis applies them to recovered Dataset B executions. No candidate is selected in this document.

1 Why process selection is the risk

LIT The most consequential finding in the literature is not a criterion but a failure mode. Two independent systematic reviews report that choosing the wrong process is the leading documented cause of RPA project failure: *"poor process selection remains consistently identified as a leading cause of RPA project failure"* [7], and *"the selection of unsuitable RPA use cases has been identified as a principal contributor to failure"* [3].

INF The ordering we propose in Step 2 therefore has to be defensible on characteristics that evidence links to success — not on whichever process happens to consume the most observed time.

2 Japanese evidence

2.1 The strongest source is a direct domain match

JP Japan's Ministry of Internal Affairs and Communications (総務省) publishes an official RPA guidebook for local governments [8]. It is a primary government source, and the processes it documents overlap closely with the process categories visible in Dataset B.

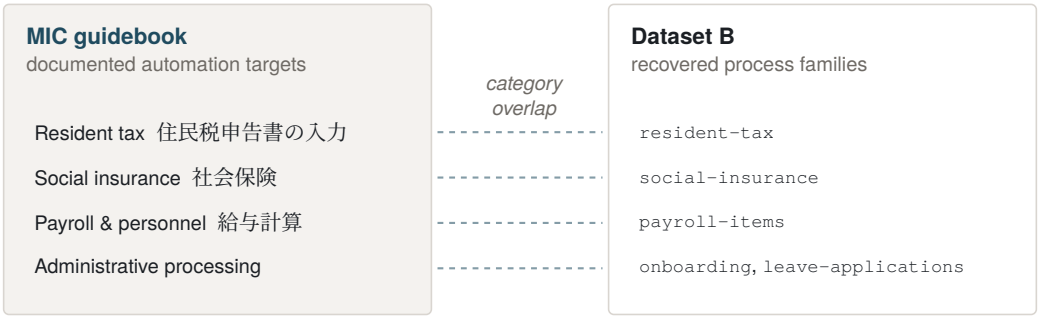


Figure 2. Category overlap between the MIC guidebook's documented automation targets and the process families Step 1 recovered from Dataset B. **This establishes only that these process *categories* are externally documented automation targets.** It is not evidence that our specific Dataset B processes should be automated, and it is not used as such.

2.2 Reported outcomes

JP The guidebook reports annual hours saved with the municipality named and the number of processes stated. We report these exactly as the source reports them.

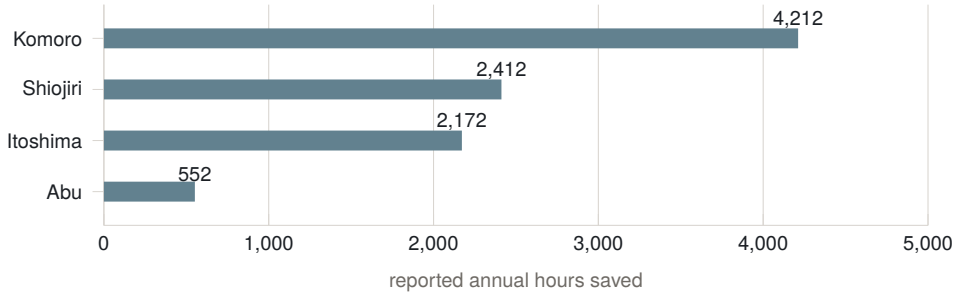


Figure 3. Reported annual hours saved in MIC guidebook case studies — **not estimates for Dataset B**. Processes automated and population: Komoro 1 process (pop. 82,102); Shiojiri 2 (65,623); Itoshima 5 (103,833); Abu 5 (28,318). The source does not convert hours to money and neither do we.

2.3 Stated selection criteria

JP The guidebook's judgement standards are, in effect, a government-published version of the academic criteria:

- 定型的で複雑な判断を要しない業務 — routine work requiring no complex judgement;
- large-volume, repetitive processing;
- an established *electronic* data input source;
- a *stable* processing target that does not keep changing;
- high overtime load in the department concerned.

> **Four of those five are properties of the work; one is volume.** INF That ratio is itself a finding, and it is the clearest external support for treating frequency as a sizing factor rather than as a qualifying one.

2.4 Stated cautions

JP The same guide names prerequisites that constrain feasibility: pair an RPA specialist with someone who knows the business process; account for **network separation** (マイナンバー / LGWAN / internet segments need separate deployments, which multiplies cost in Japanese public bodies); plan scenario maintainability and handover; and note that success requires 業務の整理・見直し — reviewing and tidying the process — not merely deploying a robot on top of it.

JP A related Japan-specific governance risk is the 野良ロボット ("stray robot") problem, documented by Mitsubishi Research Institute [9]: RPA whose owner becomes unknown continues to send mail and manipulate files unsupervised, and in reported instances a company *"remained unaware for an extended period that calculation results were wrong, which could have caused management to misjudge."* MRI recommends implementation and operating standards comparable to the internal controls applied to spreadsheets. No quantitative failure rate is given.

2.5 Private-sector evidence is weaker, and we say so

JP Japanese SI vendors publish departmental examples — attendance aggregation, monthly working-hour calculation, paid-leave balances, social-insurance change documents for employee relocations, invoice and voucher entry, CSV payment imports, payment reconciliation [10]. These describe Excel-heavy work and explicitly frame RPA as operating "standard screens" to achieve data linkage between systems at low cost where modifying the systems directly was cost-prohibitive.

Search limitation, stated explicitly

We could not find company-level Japanese case studies that identify the customer, the process, the volume *and* a disclosed outcome to a standard we would rely on. Vendor columns name processes but not customers or numbers; pages that name customers rarely disclose method. Japanese private-sector material is therefore used here only to establish *which processes are commonly targeted* — never to support an outcome or ROI claim. The MIC guidebook is the only Japanese source from which we quote outcomes.

JP Japan's structural labour shortage is the standing driver for back-office automation [14, 15, 16]. We treat this as context for *why* the client cares, not as evidence for *which* process to automate.

3 Academic evidence

3.1 The criteria that recur

LIT Across the reviews a consistent core emerges: rule-based / deterministic nature, high transaction volume, structured and digitally accessible input data, stable inputs, and low cognitive-judgement requirement, with standardisation, degree of human involvement, documentation quality and accuracy-criticality as secondary, domain-validated factors [7]. Method development runs from informal checklists through Delphi-weighted multi-criteria approaches to integer programming and time-driven activity-based costing; Wanner et al. [4, 5] is the canonical quantifiable-selection work and AHP/TOPSIS [6] the representative multi-criteria method.

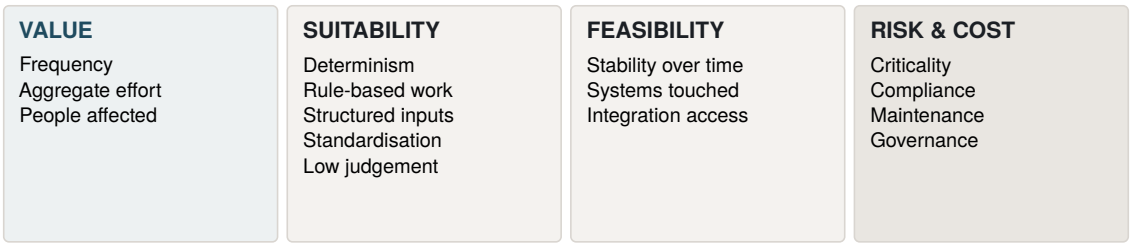


Figure 4. The recurring criteria, grouped. The grouping matters more than the list: value and suitability answer different questions and must not be summed into one number.



INF These two distinctions are the spine of the Step 2 design. The first separates gating from ranking; the second keeps suitability and value on separate axes.

3.2 The definition stated over our exact data type

LIT Bosco, Augusto, Dumas, La Rosa and Fortino [1] define automatability *for UI logs*:

> *"a routine is automatable when its first action is always triggered when a condition is met (the routine's activation condition), and the value of each parameter of each action can be computed from the values of parameters of previous actions."*

An action is *deterministic* when each parameter is either constant across executions or derived from earlier parameters — by direct copy, by a substring or concatenation transformation, or by a conditional substitution rule. The authors state two limitations: activation conditions following loop exits are not discovered, and unrelated events fracture discovered routines.

Why this matters for our data

INF This is the most useful criterion we found, for two reasons: it is defined over exactly the data type we have, and it is *computable rather than elicited*. Execution-to-execution structural repeatability within a process is our **operationalisation** of it, and Step 1's identity machinery already computes execution-to-execution similarity — applied *within* a process rather than *between* processes, the same measure becomes a determinism signal.

The distinction must be kept. The paper defines automatability in terms of activation condi-

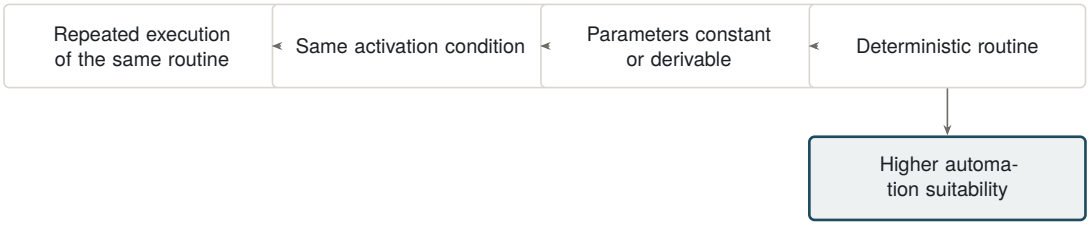


Figure 5. The automatability condition of Bosco et al. [1], stated over user interaction logs. Each link is a condition the paper defines, not a heuristic we introduced.

tions and parameter derivability. Structural similarity is a proxy we infer from it. It is not a validated production automation predictor, and this document does not claim it is.

LIT The wider framing is Leno et al. [2], whose robotic process mining pipeline our Step 1 follows.

3.3 The tensions, stated rather than resolved

These are real disagreements in the evidence, and they shape the design.

Tension	What the evidence says	Consequence for us
Frequency vs automability	Volume appears in nearly every framework, yet MIC lists it once against four work-property criteria, and UiPath's own vetting score omits volume entirely	Frequency sizes the prize; it must never *be* the ranking
Complexity: cost or value?	Cross-system complexity raises implementation cost — but it is often precisely why a human is doing the work	Report complexity twice: as value evidence and as cost. Never net them
Variability: exceptions or variants?	High variance may mean unhandleable exceptions, or legitimate handling patterns the brief explicitly asks us to identify	Inspect variants before penalising variance
Human involvement	Judgement-heavy work may still suit human-in-the-loop automation	Judgement routes modality; it does not disqualify
UI automation vs integration	Japanese vendors frame RPA as a substitute for integration where changing systems is prohibitive; the literature notes protocol and UI changes force rework	UI automation can be right *and* carry the higher maintenance burden. Say both
Automatable ≠ worth it	A process can satisfy every suitability test and still be too small to matter	Suitability and value are separate axes, not one score
Automating the as-is process	MIC: success requires 業務の整理・見直し, not RPA alone	"Redesign" must remain a permitted recommendation

4 The existing tool landscape

We reviewed the primary documentation of the category leaders rather than compiling a feature catalogue. Two findings matter.

VENDOR **Task mining does not rank candidates.** UiPath Task Mining's own documentation states it does not produce "a ranked list of automation recommendations", and that "determining automation potential and submitting a candidate to Automation Hub remains a manual step performed by a Business Analyst or SME" [13]. It captures clicks, keystrokes and hotkeys, merges variants into task graphs, and surfaces variants, step statistics and time-per-action as *evidence for a human*.

VENDOR **The vendor's candidate score is pure human judgement.** UiPath Automation Hub's Idea Score is an arithmetic average of five factors, each a 1–5 ranking supplied by the submitter: process rules, input digitisation, data structure, likelihood of process change, and documentation. Factors 1–3 form a *suitability* sub-score, 4–5 a *readiness* sub-score. The documentation states the algorithm exists "to vet out ideas that should not be automated", and that the system applies no external data sources [12]. **Volume does not appear in it at all.**

Capability	Process mining Celonis, Signavio, IBM	Task mining UiPath, AA	Intake scoring Automation Hub	Ours
Observes	system event logs with case ids	desktop UI events	human submissions	desktop UI events, no case ids
Discovers processes	yes, from case id	partially; task graphs	no	yes — inferred boundaries and identity
Ranks candidates	throughput / conformance driven	**no**	yes, from Likert input	to be designed
Estimates ROI	with supplied cost data	no	no	no — no cost data
Assesses suitability	limited	evidence only	human judgement	**from telemetry**
Needs what we lack	case ids, cost rates, org data	labelled task boundaries	an expert panel	—

INF Our position is narrow but genuine: **the industry baseline collects telemetry and then asks a human to score suitability subjectively; we are deriving suitability evidence from the telemetry itself.** This is not a claim of superiority — the human has business context we do not, and we make no comparative performance claim against any vendor product. It is a statement about where our Step 1 output can contribute.

5 Criteria carried into Step 2

Condensed to the smallest defensible set. *Role* distinguishes a hard gate from a ranking factor; gates are set in bold and are not tradeable against volume.

#	Criterion	Category	Role	Evidence
1	Determinism / repeatability	Suitability	**GATE**	LIT Bosco; JP MIC; VENDOR process rules
2	Volume / frequency	Value	RANK	LIT all reviews; JP MIC
3	Aggregate observed effort	Value	RANK	LIT; JP overtime criterion
4	Structured, digital input	Suitability	**GATE**	JP MIC; VENDOR input digitisation
5	Standardisation / variants	Suitability	RANK	LIT reviews
6	Human-judgement requirement	Suitability	**GATE**	LIT; JP MIC
7	Stability of the target	Feasibility	RANK	JP MIC; VENDOR process changes
8	Cross-system transitions	Feasibility, Value	RANK	JP Hitachi; LIT
9	Manual transfer signature	Value, Suitability	RANK	LIT data transformations; JP
10	People affected	Value	RANK	LIT
11	Maintenance & governance	Cost, Risk	RANK	JP 野良ロボット; LIT Syed
12	Criticality / compliance	Risk	**GATE** *not observable*	LIT; JP
13	Evidence sufficiency	Confidence	**GATE**	INF

Criterion 12 is a gate we cannot evaluate from telemetry. It is listed because omitting it would misrepresent the framework; it becomes a client validation question (§7).

6 Telemetry coverage

DATA Dataset B as recovered by the frozen Step 1 pipeline: **183 executions, 13 labels, 15 sessions, 127.4 minutes** of segmented activity. Step 1's measured quality on Dataset A (which has ground truth): boundary F1 **0.690** at $\tau = 2$ s against a reachability ceiling of **0.900**; identity ARI **0.681** at match rate **0.760**.

> These figures are **measurement limitations**, not accuracy claims about any automation recommendation. Dataset B has no ground truth; nothing downstream of it can be validated.

Legend. ● direct ◐ proxy or derived ○ missing

	Criterion	How measured	Confidence
●	Frequency	executions per label	high
●	Aggregate effort (*relative only*)	summed durations; the brief warns waiting time is compressed	medium
●	People affected	operator hash per execution	medium
●	Systems / surfaces touched	surfaces and origins per execution	high
●	Cross-system transitions	execution-graph transitions	high
●	Interaction composition	typing / click / form-input / shortcut mix	high
●	Manual transfer signature	clipboard events per execution	high
●	Duration variance	per-label distribution	medium
●	Fragmentation / interruption	Step 1 execution fragments	medium
◐	Variant count	sub-labels within a route family	medium
◐	**Determinism**	within-process execution similarity	medium–high
◐	Input structure	form-field interaction vs free typing	low–medium
◐	Human judgement	dwell without input; decision-point dispersion	**low**
○	Environment stability	single capture window	—
○	Labour cost / FTE rate	—	—
○	Business criticality	—	—
○	Compliance exposure	inferable from process type only	—
○	API availability	—	—
○	Error / rework rate	—	—
○	True volume & seasonality	15 sessions, one window	—

7 Missing evidence becomes declared assumptions

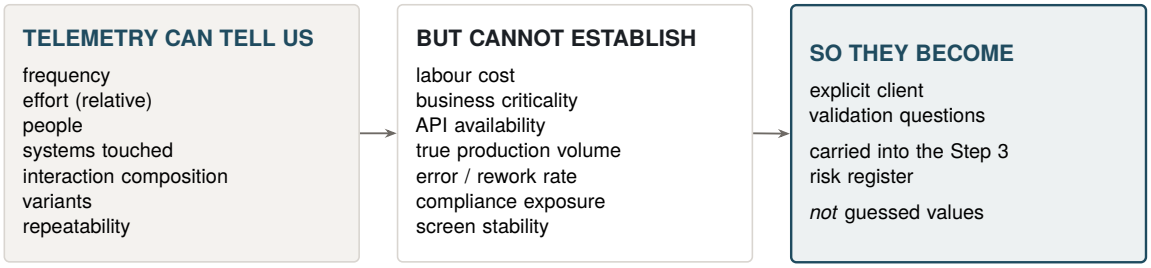


Figure 6. Missing is not ignored. Each unobservable criterion converts into a question for the client rather than an invented number — which is also what the assignment asks us to anticipate from limited information.

INF The validation questions that follow from the table above:

1. What is the fully loaded hourly cost of the staff performing this work?
2. What is the true annual volume, and how seasonal is it? Our window is roughly two hours of activity.
3. Do the target systems expose an API or a bulk import, making UI automation unnecessary?
4. What is the error and rework rate today, and what does a wrong record cost downstream?
5. What statutory or audit obligations attach to this process?
6. How often do these screens change, and who owns them?
7. Who would own and maintain the automation? The 野良ロボット risk is a governance question, not a technical one.

8 Automation modality

INF, grounded in LIT and JP. Modality is a routing decision, not a consequence of rank: the highest-priority process does not automatically warrant a UI robot.

Observed characteristics	Indicated modality
High repeatability, deterministic parameters, structured fields, stable screens, no API	**Deterministic UI automation (RPA)**
As above, *but* the target exposes an API or bulk import	**API / integration** — preferred; lower maintenance than UI automation
Multiple systems, human acting as the integration layer, handoffs between people	**Workflow orchestration**
Volume concentrated in extracting values from unstructured documents	**Document / IDP**
Repeatable skeleton with a judgement step inside it	**AI-assisted or human-in-the-loop**
High variance that inspection shows to be genuine variants rather than noise	**AI-assisted**, or redesign first
Complexity exists only because systems do not talk to each other	**Process redesign / integration** — MIC's 業務の整理・見直し
Low volume, high judgement, high risk	**Do not automate**

JP The Japanese vendor framing is important and honest: RPA is chosen as a *substitute for integration* when changing the systems is cost-prohibitive [10]. That is a legitimate reason, and it carries a known cost — protocol and UI changes force bot rework [3]. Both halves belong in the recommendation.

9 Choosing a prioritisation method

Method	Problem in our setting	Decision
Single weighted score	needs weights we cannot elicit; hides the trade-offs	**reject**
Frequency × effort	measurable, but contradicts the evidence that suitability dominates	reject as ranking; keep as the value axis
Weighted MCDA (AHP / TOPSIS)	requires expert pairwise comparison we have no access to	**reject** — cannot be honestly instantiated
Hard gates then ranking	— matches MIC's criteria and the "vet out" design of vendor intake	**adopt**
Pareto frontier	no false precision, but gives no single answer, and Step 3 needs one target	partial
Impact × feasibility matrix	management-legible; separates the two axes the evidence separates	**adopt as presentation**
Two-stage discovery then feasibility	mirrors both MIC and vendor intake practice	**adopt as structure**

Evidence resolution should determine reporting resolution

What we have

183 executions
15 sessions
compressed waiting times
no labour cost data
no business criticality
Step 1 identity at ARI 0.681

Therefore

Do *not* report: `candidate score = 0.73`

Prefer: **High** / **Medium** / **Low**, each with the evidence behind it and a stated confidence.

INF A number we cannot defend to the second decimal should not be written to the second decimal. Ordinal bands with visible evidence are more honest, and more useful to a manager, than a composite score whose weights were invented.

10 Design principles for Step 2

1. **Gate first, rank second.** Determinism, structured input, judgement and evidence sufficiency are gates. Frequency and effort rank what survives.
2. **Never rank on frequency alone.** It sizes the prize; it does not indicate automability.
3. **Keep value and suitability on separate axes** and present them as a matrix.
4. **Route modality explicitly**, and justify why the alternatives were rejected.
5. **No monetary ROI.** No cost data exists, and the brief warns the timings are compressed. Compare processes against each other, as it instructs.

6. **Attach a confidence level to every candidate**, derived from executions observed, sessions and operators covered, and Step 1's own measured uncertainty.
7. **Every missing dimension becomes a written validation question**, not a guessed value.
8. **"Redesign" and "do not automate" must remain reachable conclusions**, or the analysis is not a real assessment.
9. **Exclude non-process overhead.** DATA Portal-root navigation appears as a recovered label but is not business work.
10. **Inspect variants before penalising variance.** The brief explicitly asks whether different handling patterns exist within a process.

11 What this enables next

This research does not select the automation candidate. It establishes the evidence framework and the boundary of what our telemetry can support.

The next step applies the framework to every recovered Dataset B process: build the descriptive inventory the brief asks for, compute the candidate-level evidence, apply the hard gates, assess impact against suitability and feasibility, expose the missing evidence as validation questions, and only then select the process offering the strongest defensible combination of value and feasibility — together with the modality appropriate to it.

INF Selecting a candidate before that analysis has run would reverse the order this document exists to establish.

Substance source of truth: `docs/STEP2_RESEARCH_REVIEW.md`. This document reformats that content; it does not extend it.

Search limitations

Japanese company-level case studies identifying customer, process, volume and a disclosed outcome together could not be found at an acceptable evidence standard; vendor columns name processes but not customers or numbers. Outcome figures in this document come only from the MIC guidebook [8]. The IPA *DX動向2025* PDFs were not parsed and no figures are quoted from them.

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